FREE GLOBAL SECTIONS ON AN AFFINE ETALE NEIGHBOURHOOD FOR A FINITE TYPE MODULE FLAT AT A POINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The affine neighbourhood statement in the finite presentation case, and freeness of a projective module over a local ring. Same chapter flat.tex, label lemma-flat-finite-presentation-affine-neighbourhood-projective (proof line 5752, claimed as S14-algebra-projective-neighbourhood); Algebra, label algebra-theorem-projective-free-over-local-ring (proof line 5754). The first provides the affine etale neighbourhood with projective global sections once the module is of finite presentation, and the second turns projectivity over the local base into freeness.
- The local structure statement giving a finitely presented pullback after an etale neighbourhood. Same chapter flat.tex, label proposition-finite-type-flat-at-point, invoked at proof line 5750. Allows the reduction to a module of finite presentation by passing to an affine neighbourhood on the base and an open of the base change.

Source reference. Original theorem 0593: sources/stacks-chow/tags.txt, source lines 6801–6801. Navigation pointer for \ref{algebra-theorem-projective-free-over-local-ring} used inside the counted span (line 5754); the argument that consumes it is retained in the span.

Source reference. Original lemma 05ME: sources/stacks-flat/pinned-original-source.tex, source lines 5662–5662. Navigation pointer for \ref{lemma-flat-finite-presentation-affine-neighbourhood-projective} used inside the counted span (line 5752); the argument that consumes it is retained in the span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original proposition 05I5: sources/stacks-flat/pinned-original-source.tex, source lines 2477–2477. Navigation pointer for \ref{proposition-finite-type-flat-at-point} used inside the counted span (line 5750); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism which is locally of finite type. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module which is of finite type. Let $x \in X$ with $s = f(x) \in S$. If $\mathcal{F}$ is flat at x over S there exists an affine elementary étale neighbourhood $(S', s') \to (S, s)$ and an affine open $U' \subset X \times_S \mathrm{Spec}(\mathcal{O}_{S',s'})$ which contains $x' = (x, s')$ such that $\Gamma(U', \mathcal{F}|_{U'})$ is a free $\mathcal{O}_{S',s'}$-module.*

Proof. The question is Zariski local on X and S. Hence we may assume that X and S are affine. Then we can find a closed immersion $i : X \to \mathbf{A}^n_S$ over S. It is clear that it suffices to prove the lemma for the sheaf $i_*\mathcal{F}$ on $\mathbf{A}^n_S$ and the point $i(x)$. In this way we reduce to the case where $X \to S$ is of finite presentation. After replacing S by $\mathrm{Spec}(\mathcal{O}_{S',s'})$ and X by an open of $X \times_S \mathrm{Spec}(\mathcal{O}_{S',s'})$ we may assume that $\mathcal{F}$ is of finite presentation, see Proposition 05I5. In this case we may appeal to Lemma 05ME and Algebra, Theorem 0593 to conclude. □